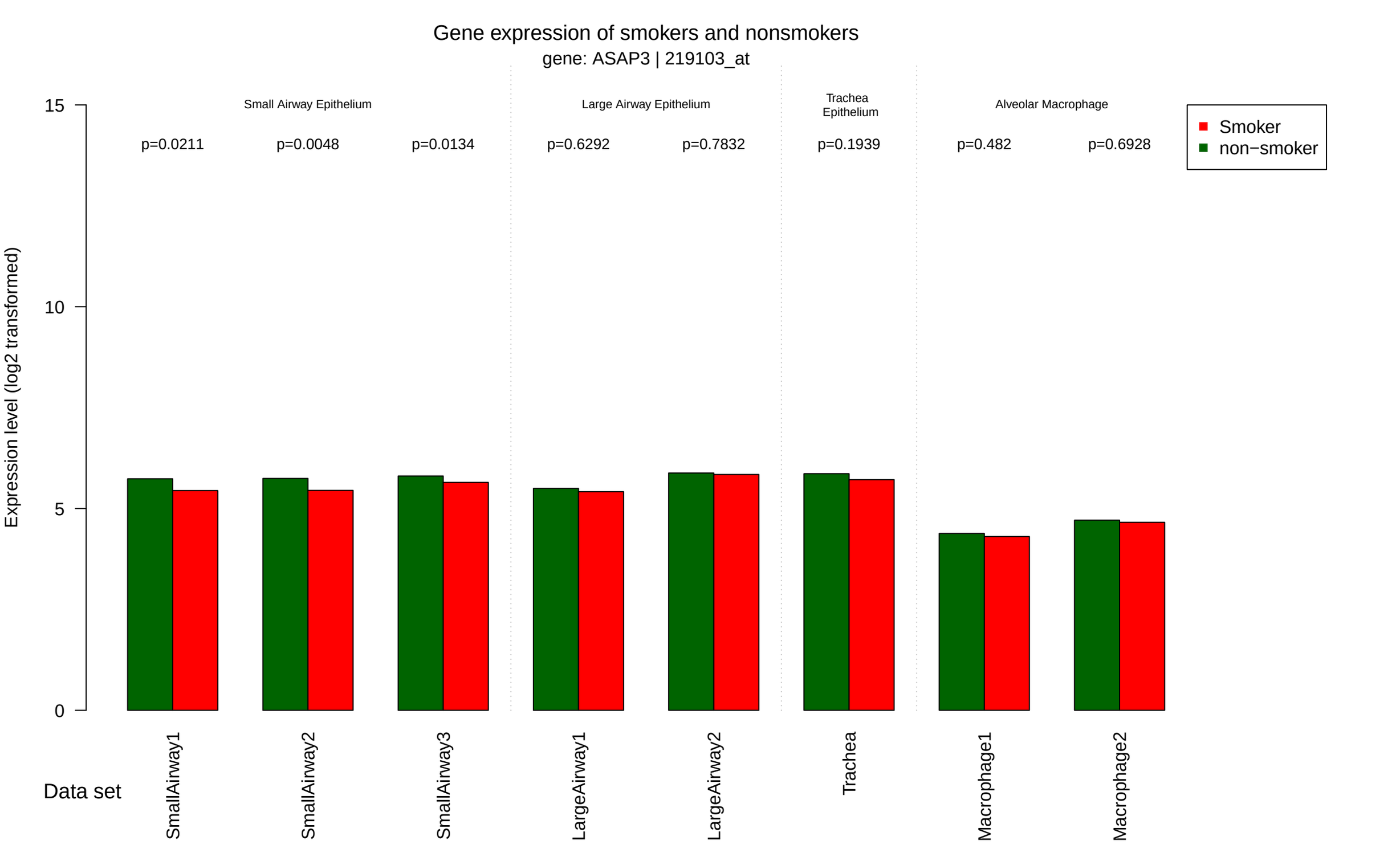
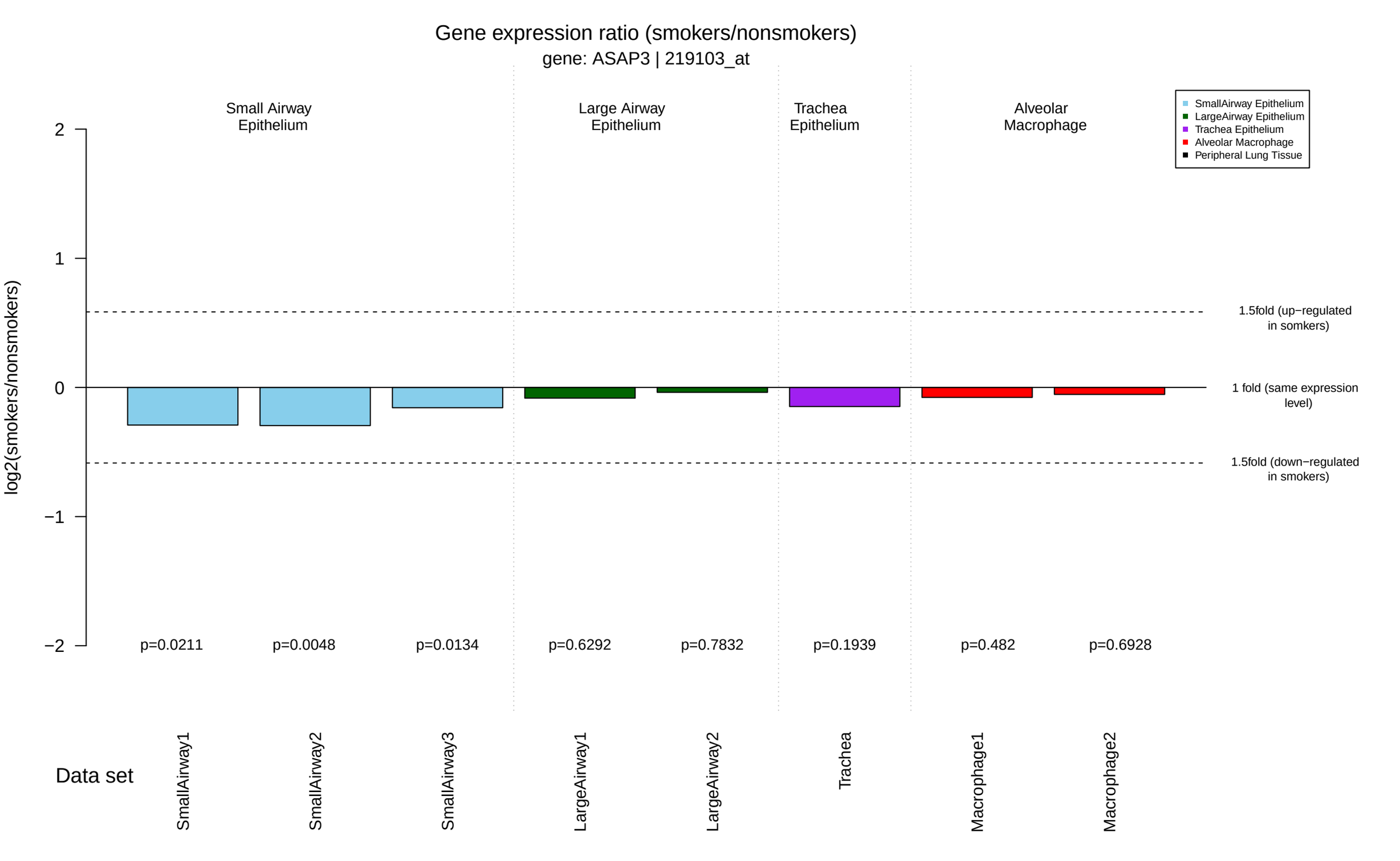
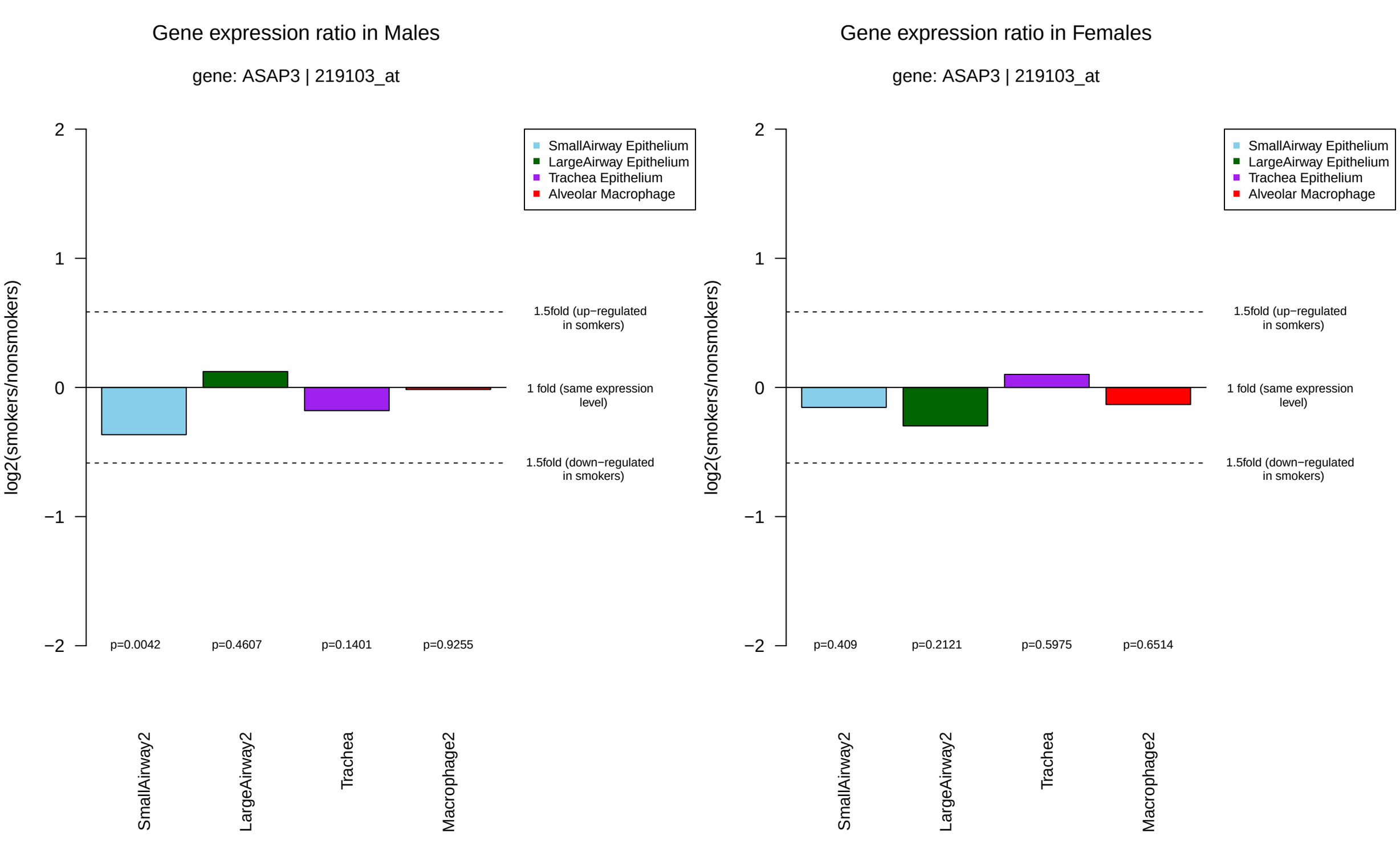
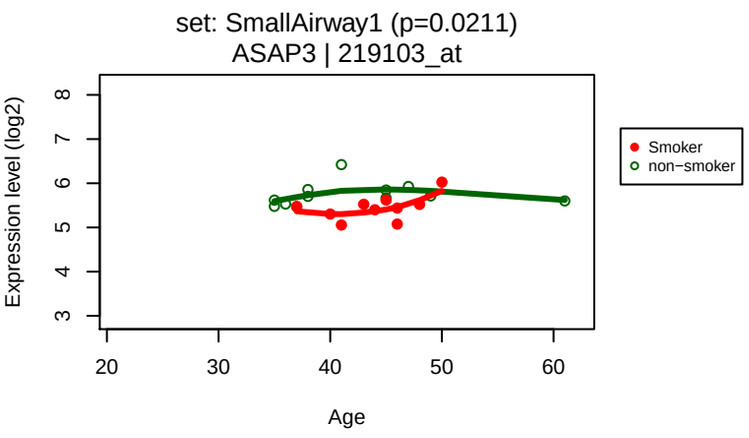
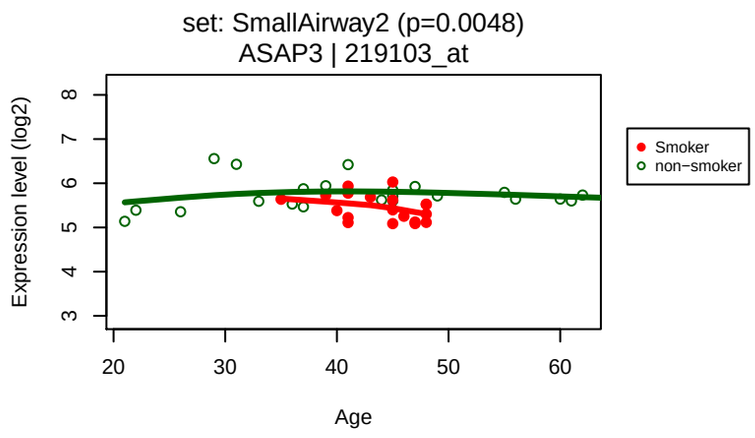
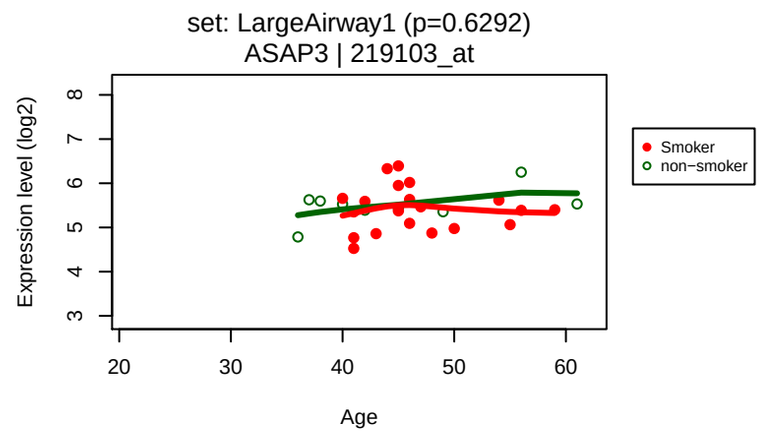

Gene expression of smokers and nonsmokers
gene: ASAP3 | 219103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0211
p=0.0048
p=0.0134
p=0.6292
p=0.7832
p=0.1939
p=0.482
p=0.6928
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

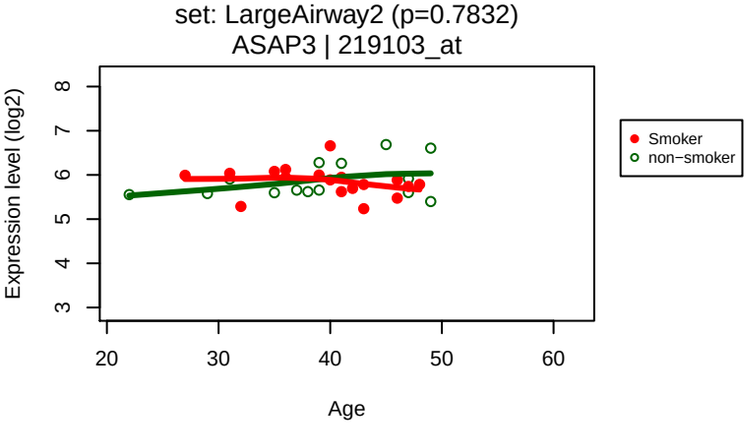

Gene expression ratio (smokers/nonsmokers)
gene: ASAP3 | 219103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0211
p=0.0048
p=0.0134
p=0.6292
p=0.7832
p=0.1939
p=0.482
p=0.6928
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

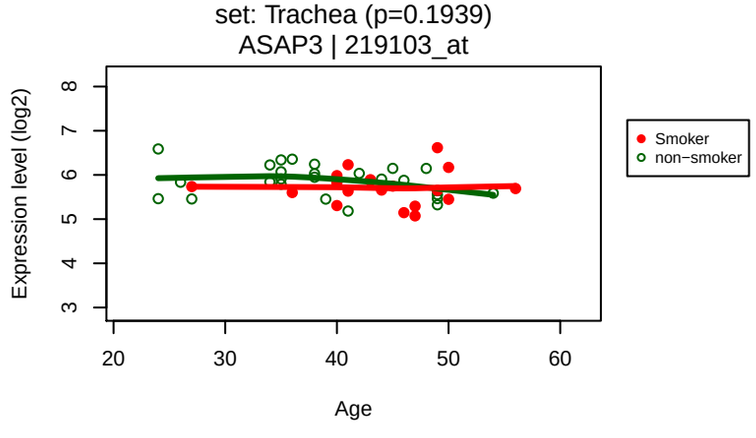

Gene expression ratio in Males
gene: ASAP3 | 219103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0042
p=0.4607
p=0.1401
p=0.9255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAP3 | 219103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.409
p=0.2121
p=0.5975
p=0.6514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

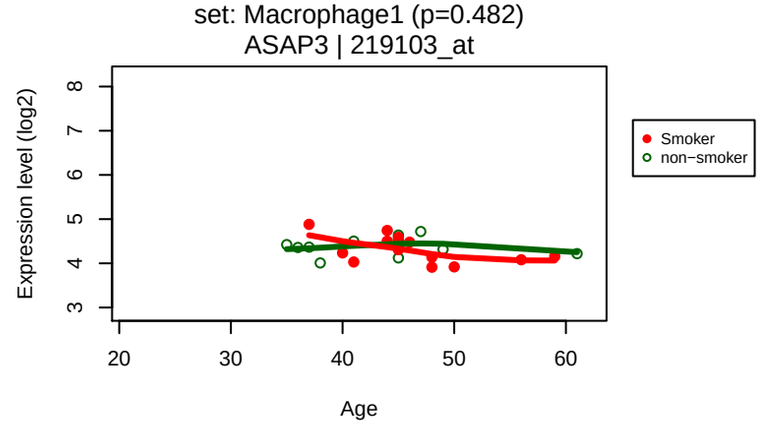

set: SmallAirway1 (p=0.0211)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker

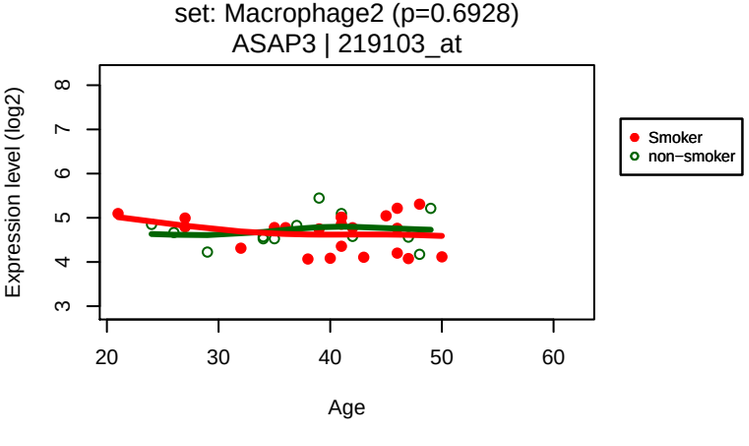

set: SmallAirway2 (p=0.0048)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker

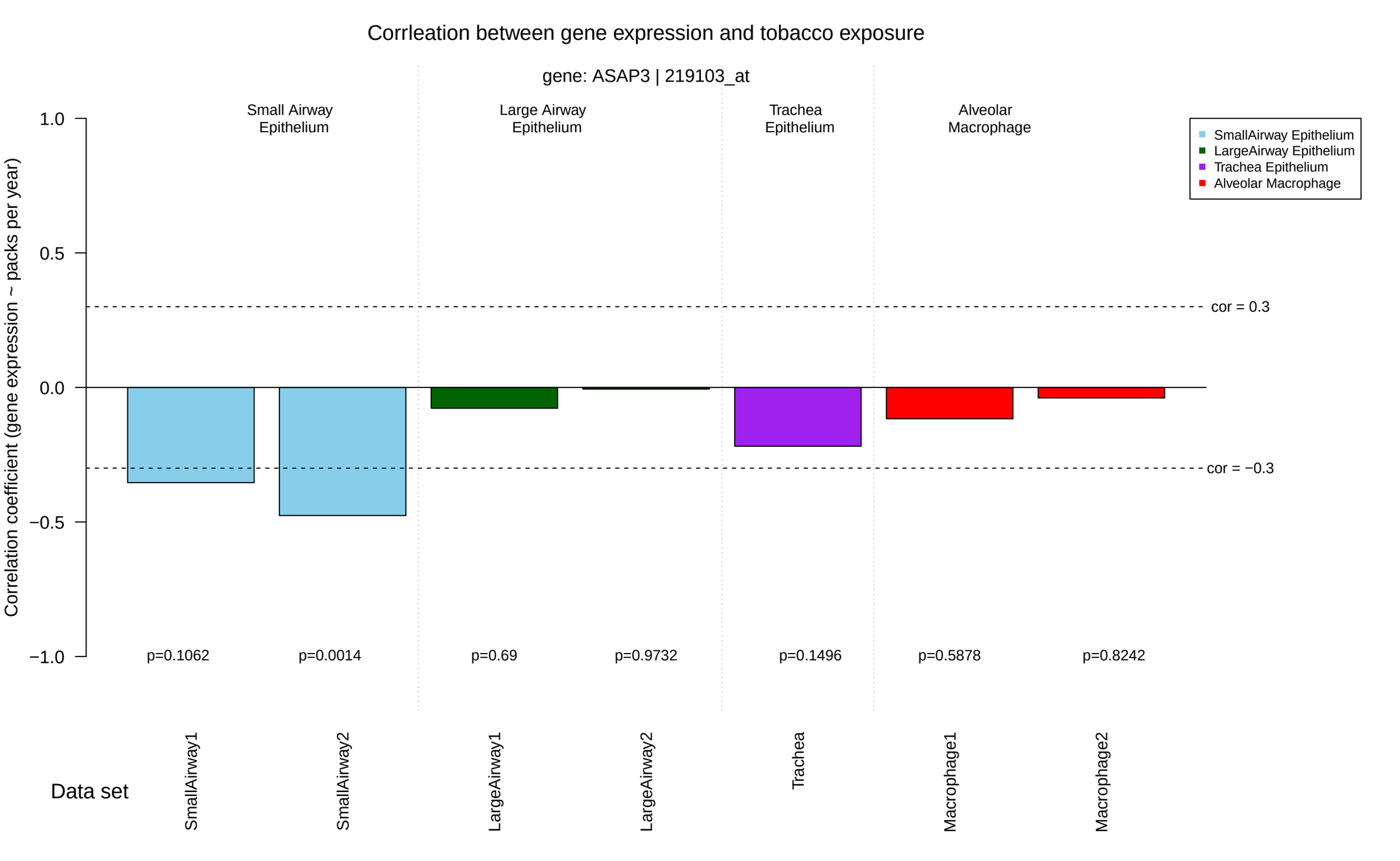

set: LargeAirway1 (p=0.6292)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7832)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1939)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.482)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6928)
ASAP3 | 219103_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAP3 | 219103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1062
p=0.0014
p=0.69
p=0.9732
p=0.1496
p=0.5878
p=0.8242
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2